

Corollary 8.17 (Euler-Poincaré formula).

For every d -dimensional polytope,

$$f_0 - f_1 + \cdots + (-1)^{d-1} f_{d-1} = 1 - (-1)^d.$$

Proof. In order to do induction, we need to consider the alternating sum of face numbers for general polytopal complexes. For this we define the (*reduced*) Euler characteristic $\chi(\mathcal{D})$ of a polyhedral complex of dimension at most d , by

$$\chi(\mathcal{D}) := -f_{-1} + f_0 - f_1 + \cdots + (-1)^d f_d.$$

Now if $\mathcal{D}$ and $\mathcal{D}'$ are polytopal complexes such that the union is a polytopal complex too (that is, if $F \cap F' \in \mathcal{D} \cap \mathcal{D}'$ for $F \in \mathcal{D}$, $F' \in \mathcal{D}'$), then the Euler characteristic is additive:

$$\chi(\mathcal{D}) + \chi(\mathcal{D}') = \chi(\mathcal{D} \cup \mathcal{D}') + \chi(\mathcal{D} \cap \mathcal{D}').$$

We prove by induction on d that the complex of a polytope $\mathcal{C}(P)$ always has Euler characteristic 0, and thus its boundary has Euler characteristic $(-1)^{d-1}$:

$$\chi(\mathcal{C}(P)) = 0, \quad \chi(\mathcal{C}(\partial P)) = (-1)^{d-1}.$$

This is clear for $d \leq 1$. Now if P is a d -polytope with shelling order $F_1, F_2, \dots$, then we have more precisely that

$$\chi(\mathcal{C}(F_1 \cup F_2 \cup \cdots \cup F_j)) = \begin{cases} 0 & \text{for } 1 \leq j < f_{d-1} \\ (-1)^{d-1} & \text{for } j = f_{d-1} \end{cases}$$

— which follows by induction on j and dimension, since the facets F_j that we add in are $(d-1)$ -polytopes, the Euler characteristic is additive, and the intersection $F_j \cap (\bigcup_{i < j} F_i)$ is a shellable part of, but not the whole boundary of F_j , for $j < f_{d-1}$. This last fact is immediate from Lemma 8.10, or geometrically in the special case of a line shelling. $\square$

8.3 h -Vectors and Dehn-Sommerville Equations

From now on, to the end of this lecture, all polytopes are simplicial.

For simplicial polytopes, the combinatorics of shellings can be described even more concretely. For the following let P be a simplicial d -polytope, so its boundary is a simplicial complex $\mathcal{C} := \mathcal{C}(\partial P)$ of dimension $d-1$. Let $V := \text{vert}(P)$ be the vertex set of P of size $n := f_0(P)$.

We identify the proper faces of P with their vertex sets — that is, we identify the “geometric simplicial complex $\mathcal{C}$ ” with the “abstract simplicial

complex” on the finite set V . In particular, the facets of P are $(d-1)$ -simplices, and thus correspond to d -subsets of V . The complex $\mathcal{C}$ is pure $(d-1)$ -dimensional, so it is completely determined by the family of facets $\mathcal{F} \subseteq \binom{V}{d}$. All the other faces in $\mathcal{C}$ correspond to the subsets of facets in $\mathcal{F}$.

Now fix a shelling order $F_1, F_2, \dots$ on the facets in $\mathcal{F}$. We define the *restriction* R_j of the face F_j as the set of all vertices $v \in F_j$ such that $F_j \setminus v$ is contained in one of the earlier facets:

$$R_j := \{v \in F_j : F_j \setminus v \subseteq F_i \text{ for some } 1 \leq i < j\}.$$

The main observation here is that when we build up $\mathcal{C}$ according to the shelling, the *new faces* at the j th step are exactly the vertex sets G with

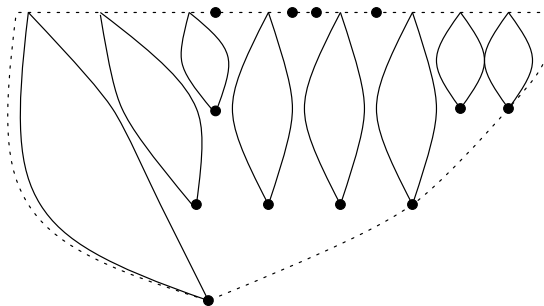
$$R_j \subseteq G \subseteq F_j.$$

In fact, a face G that is new is necessarily a subset of F_j : if it misses a vertex $v \in R_j$, then it was already contained in a previous facet, by construction. Finally, if G satisfies $R_j \subseteq G \subseteq F_j$ but is not new, with $G \subseteq F_i$ for some $i < j$, then by the definition of shellings G is contained in some F_l ($l < j$) such that $F_l \cap F_j = F_j \setminus w$ is a facet of F_j . From $F_j \setminus w \subseteq F_l$ we get $w \in R_j$, and from $R_j \subseteq G \subseteq F_l \cap F_j = F_j \setminus w$ we get $w \notin R_j$: a contradiction.

Thus every shelling gives us a partition $I_1 \uplus \dots \uplus I_s$ of the set of faces of the simplicial complex into intervals of the form

$$I_j := \{G : R_j \subseteq G \subseteq F_j\}.$$

A pure simplicial $(d-1)$ -complex that has such a partition (with exactly one part for each facet of $\mathcal{C}$) is called *partitionable*, a concept developed independently by Provan [447, App. 4], Stanley [513, p.149], and Garsia [94, p. 607]. Shellable simplicial complexes are partitionable, as we have seen. Our drawing tries to illustrate how the face poset of a partitionable complex decomposes into intervals.



For a partitionable simplicial complex the f -vector can be read off from the partition. Namely, if $|R_j| = i$, then there are exactly $\binom{d-i}{k-i}$ $(k-1)$ -faces

contained in the part I_j , and thus

$$f_{k-1} = \sum_{j=1}^s \binom{d - |R_j|}{k - |R_j|}.$$

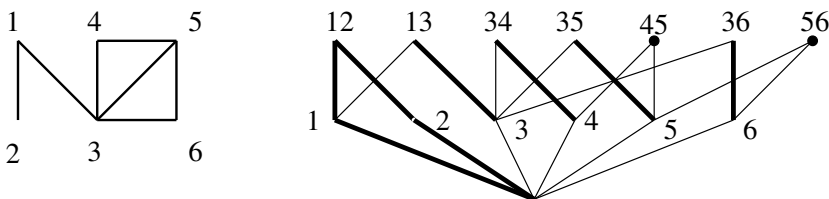
Let $h_i = h_i(\mathcal{C})$ denote the number of parts in the partition such that the corresponding restriction set has size i :

$$h_i(\mathcal{C}) := \left| \{j : |R_j| = i, 1 \leq j \leq s\} \right|.$$

The h -vector of a partitionable simplicial $(d-1)$ -complex $\mathcal{C}$ is this sequence

$$\mathbf{h}(\mathcal{C}) = (h_0, h_1, \dots, h_d).$$

For example, the following graph (1-dimensional complex) $\mathcal{C}$ on 6 vertices has $\mathbf{f} = (1, 6, 7)$.



It is connected, hence shellable, a shelling order being given by the facet ordering

$$12, 13, 34, 35, 45, 36, 56.$$

The bold edges in the face poset indicate the corresponding partition. Its “minimal new faces” are

$$\emptyset, 3, 4, 5, 45, 6, 56,$$

and thus we get a contribution of “1” in this order to

$$h_0, h_1, h_1, h_1, h_2, h_1, h_2,$$

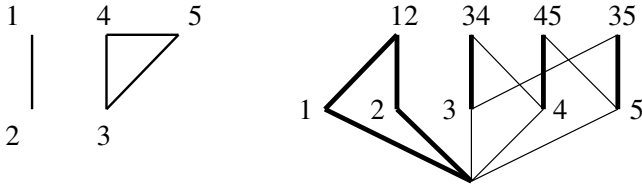
and thus

$$\mathbf{h}(\mathcal{C}) = (1, 4, 2).$$

All partitionable complexes have nonnegative h -vectors, since h_i is the number of minimal faces in the partition that have i vertices.

We have seen that shellable simplicial complexes are partitionable. The converse is not true: here is a partitionable but nonshellable complex, with

$d = 2$, $n = 5$, $\mathbf{f} = (1, 5, 4)$ and $\mathbf{h} = (1, 3, 0)$.



The f -vector of the shellable simplicial complex $\mathcal{C}$ can be computed from the h -vector. In fact, adding up the contributions at the individual steps of the shelling gives

$$\begin{aligned} f_{k-1} &= \sum_{j=1}^s \binom{d - |R_j|}{k - |R_j|} \\ &= \sum_{i=0}^k h_i \binom{d-i}{k-i} \\ &= h_k + (d-k+1)h_{k-1} + \cdots + \binom{d-1}{k-1}h_1 + \binom{d}{k}h_0. \end{aligned}$$

However, the f -vector also determines the h -vector: from this formula we can recursively compute h_k from f_{k-1} together with $(h_0, \dots, h_{k-1})$. Here is one way to do the bookkeeping. We consider the f -polynomial

$$\mathbf{f}(x) := f_{d-1} + f_{d-2}x + \cdots + f_0x^{d-1} + f_{-1}x^d = \sum_{i=0}^d f_{i-1}x^{d-i}$$

and the h -polynomial

$$\mathbf{h}(x) := h_d + h_{d-1}x + \cdots + h_1x^{d-1} + h_0x^d = \sum_{i=0}^d h_ix^{d-i}.$$

From the above derivation, we see that a shelling step with $|R_j| = i$ contributes a summand of $(x+1)^{d-i}$ to the f -polynomial. Thus, we get a formula

$$\mathbf{f}(x) = \sum_{i=0}^d h_i(x+1)^{d-i} = \mathbf{h}(x+1).$$

If we compare the coefficients of x^{d-k} in this formula, then we get the above expression of f_{k-1} in terms of the h_i . However, $\mathbf{f}(x) = \mathbf{h}(x+1)$, so we certainly also have

$$\mathbf{h}(x) = \mathbf{f}(x-1).$$

Now if we compare the coefficients of x^{d-k} on both sides of this equation, we get a formula for h_k in terms of the f_i . We take the result, and make it into a definition.

Definition 8.18. Let $\mathcal{C}$ be a simplicial complex of dimension $d-1$. The h -vector of $\mathcal{C}$ is

$$\mathbf{h}(\mathcal{C}) = (h_0, h_1, \dots, h_d) \in \mathbb{Z}^{d+1},$$

given by the formula

$$h_k := \sum_{i=0}^k (-1)^{k-i} \binom{d-i}{d-k} f_{i-1},$$

that is,

$$\begin{aligned} h_k &= \sum_{i=0}^k (-1)^{k-i} \binom{d-i}{d-k} f_{i-1} \\ &= f_{k-1} - (d-k+1)f_{k-2} + \binom{d-k+2}{2}f_{k-3} - \dots \\ &\quad \dots + (-1)^{k-1}f_0 \binom{d-1}{k-1} + (-1)^k \binom{d}{k}. \end{aligned}$$

In particular, we have $h_0 = 1$, $h_1 = f_0 - d$, and

$$h_d = f_{d-1} - f_{d-2} + f_{d-3} - \dots + (-1)^{d-1}f_0 + (-1)^d.$$

Also, it is easy to verify $h_0 + h_1 + \dots + h_d = f_{d-1}$, which for partitionable complexes holds by construction — this is just $\mathbf{f}(0) = \mathbf{h}(1)!$

The advantage of the definition of h -vectors in 8.18 is that it makes sense even if the simplicial complex is not partitionable, and in the case of a partitionable complex it shows that the numbers h_i are independent of the particular partition that we have chosen. Taking it as the definition, we have proved the following theorem.

Theorem 8.19. *Let $\mathcal{C}$ be a pure, d -dimensional simplicial complex. If $\mathcal{C}$ is partitionable, then the h -vector is nonnegative. If $\mathcal{C}$ is even shellable, then the entry h_i counts the facets in a shelling whose restriction has size i , and this number is independent of the particular shelling chosen.* $\square$

Instead of explicit evaluation of the formulas, the h -vector can also be computed by a difference table, a variant of Pascal's triangle, known as *Stanley's trick* [516, p. 213] (see also [356, p. 5]). For this we write the numbers f_i to the last entries of the rows of Pascal's triangle (to the place where ordinarily we would put $\binom{i+1}{i+1} = 1$), and then compute the other entries as

$$\text{upper right neighbor} - \text{upper left neighbor}.$$

Examples 8.20. We compute the h -vector for three different complexes.

- (i) For the first graph considered above, with f -vector $\mathbf{f} = (1, 6, 7)$, we get a table

$$\begin{array}{ccccccc} & & & & \mathbf{1} & & \\ & & & & 1 & & \mathbf{6} \\ & & & 1 & & 5 & & \mathbf{7} \\ \hline \mathbf{h} = (& 1 & & 4 & & 2 & &) \end{array}$$

where the entries of the f -vector in the table appear in boldface.

- (ii) Similarly, for the boundary of the octahedron C_3^Δ we have $\mathbf{f} = (1, 6, 12, 8)$, and the table reads

$$\begin{array}{cccccccc} & & & & & & \mathbf{1} & \\ & & & & & & 1 & & \mathbf{6} \\ & & & & 1 & & 5 & & \mathbf{12} \\ & & 1 & & 4 & & 7 & & \mathbf{8} \\ \hline \mathbf{h} = (& 1 & & 3 & & 3 & & 1 &) \end{array}$$

— to test this, draw an octahedron, and compute the h -vector from a shelling!

- (iii) For a quite pathological example, consider the 5-dimensional simplicial complex on 12 vertices that consists of a single 5-simplex plus 6 isolated vertices. For this we have

$$\mathbf{f} = (1, 6 + 6, \binom{6}{2}, \binom{6}{3}, \binom{6}{4}, \binom{6}{5}, 1) = (1, 12, 15, 20, 15, 6, 1),$$

and Stanley's difference table yields the following:

$$\begin{array}{ccccccccccccccc} & & & & & & & & \mathbf{1} & & & & & \\ & & & & & & & & 1 & & & \mathbf{12} & & \\ & & & & & & & 1 & & 11 & & \mathbf{15} & & \\ & & & & 1 & & 10 & & 4 & & \mathbf{20} & & \\ & & 1 & & 9 & & -6 & & 16 & & \mathbf{15} & & \\ & 1 & & 8 & & -15 & & 22 & & -1 & & \mathbf{6} & & \\ & 1 & & 7 & & -23 & & 37 & & -23 & & 7 & & \mathbf{1} \\ \hline \mathbf{h} = (& 1 & & 6 & & -30 & & 60 & & -60 & & 30 & & -6 & &) \end{array}$$

In particular, the h -vector has large negative components. This cannot happen for shellable complexes, but this one is not shellable: it is not even pure, and it is disconnected.

Why do we study h -vectors? For various problems about simplicial polytopes, h -vectors are a much more convenient and concise way to encode the information about the face numbers than f -vectors.

A first striking instance for this are the Dehn-Sommerville equations. Their history starts with the observation that for simplicial polytopes we get extra equations on the f -vectors from double-counting. In fact, for simplicial 3-polytopes we see that every edge is in two facets, while every facet has three edges. Thus we get $2f_1 = 3f_2$ from two ways of counting the edge-facet incidences. Similarly, for simplicial d -polytopes we derive

$$2f_{d-2} = d f_{d-1}.$$

This is the only new equation for $d \leq 4$, but in higher dimensions there are more (and more complicated) ones. Dehn [181] did the case $d = 5$, and the general case was done by Sommerville [508]. Here is the version in terms of the h -vector.

Theorem 8.21 (Dehn-Sommerville equations).

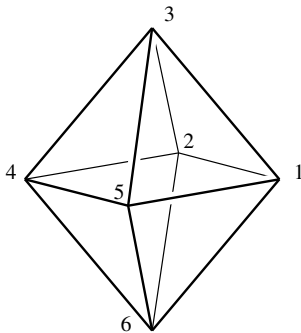
The h -vector of the boundary of a simplicial d -polytope satisfies

$$h_k = h_{d-k} \quad \text{for } k = 0, 1, \dots, d.$$

Proof. (by McMullen [389]) We use Lemma 8.10 and the observations in its proof. Namely, if $F_1, \dots, F_s$ is a shelling, then its reverse $F_s, \dots, F_1$ is a shelling as well. Furthermore, if F_i comes earlier than F_j (that is, $i < j$) in the first shelling, then it comes later in the reversed shelling. From this we see that the restriction set for F_j in the reversed shelling is exactly $F_j \setminus R_j$: the complement of the restriction set for the original shelling. Thus if F_j contributes “1” to h_k in the original shelling (where $k = |R_j|$), then it contributes “1” to h_{d-k} in the reversed shelling (where $d-k = |F_j \setminus R_j|$). Thus the value of h_k computed by the original shelling is the same as the value of h_{d-k} computed by the reversed shelling.

However, by Theorem 8.19 the h -vector is independent of the shelling chosen to compute it, and hence $h_k = h_{d-k}$. $\square$

Example 8.22. If the vertices of the octahedron are numbered as in the sketch,



then a shelling order is given by

$$123, 126, 135, 156, 234, 246, 345, 456,$$

where the corresponding minimal new faces are

$$\emptyset, 6, 5, 56, 4, 46, 45, 456,$$

and thus we get a contribution of “1” in this order to

$$h_0, h_1, h_1, h_2, h_1, h_2, h_2, h_3,$$

and thus

$$\mathbf{h}(\mathcal{C}) = (1, 3, 3, 1),$$

as we had computed in Example 8.20(ii). Now we reverse the shelling, to get the shelling order

$$456, 345, 246, 234, 156, 135, 126, 123,$$

where the corresponding minimal new faces are

$$\emptyset, 3, 2, 23, 1, 13, 12, 123,$$

and thus we get a contribution of “1” in this order to

$$h_0, h_1, h_1, h_2, h_1, h_2, h_2, h_3,$$

and from this we derive the same h -vector as before, of course.

There are various ways to write the Dehn-Sommerville equations in terms of the f -vector, ranging from the “obvious” one,

$$\sum_{i=0}^k (-1)^{k-i} \binom{d-i}{d-k} f_{i-1} = \sum_{i=0}^{d-k} (-1)^{d-k-i} \binom{d-i}{k} f_{i-1},$$

obtained by expanding the equation $h_k = h_{d-k}$ in terms of the face numbers f_i , to the perhaps most elegant and simple version,

$$f_{k-1} = \sum_{i=k}^d (-1)^{d-i} \binom{i}{k} f_{i-1}.$$

Still, $h_k = h_{d-k}$ is hard to beat in its simplicity.

In all of these versions, the equation obtained for $k = 0$ is the Euler-Poincaré formula.

The Dehn-Sommerville equations for $0 \leq k < \frac{d}{2}$ are linearly independent conditions on the h -vector (this is obvious) and thus on the f -vector (because f - and h -vectors are linearly equivalent). For the proof that these equations give a complete list of all linear relations, we refer to Grünbaum’s book [252, Sect. 9.2]. There you also find a direct proof of the various versions of the Dehn-Sommerville equations, by double-counting incidences (generalizing our argument for $2f_{d-2} = df_{d-1}$ above). However, Grünbaum did not yet have shelling available as a technique, which yields the most elegant argument.

8.4 The Upper Bound Theorem

“What is the maximal number of k -faces for a d -polytope with n vertices?”

The answer to this question is the *upper bound theorem*: “The cyclic polytope $C_d(n)$ has the maximal number of k -faces for all k .” This claim made by Motzkin [415] in 1957 became known as the *upper-bound conjecture*. During a long and involved history (see also Grünbaum [252]), including premature announcements and many partial results, it was proved for polytopes with “few” vertices (i.e., $n \leq d + 3$, by Gale [222]) and for polytopes with “many” vertices by Klee [324] (see the next section!), and in “low” dimensions (for $d \leq 8$); see Grünbaum [252, p.175]. You may observe that the result is quite easily derived from the Dehn-Sommerville equations for $d \leq 5$.

Finally, in 1970 McMullen gave a complete proof of the upper-bound conjecture — since then it has been known as the upper bound theorem. McMullen’s proof is amazingly simple and elegant, combining two key tools: shellability and h -vectors. This section presents this proof of the upper bound theorem, following McMullen’s original paper [389]. Here is the theorem.

Theorem 8.23 (Upper bound theorem). (McMullen [389])

If P is a d -polytope with $n = f_0$ vertices, then for every k it has at most as many k -faces as the corresponding cyclic polytope (cf. Example 0.6):

$$f_{k-1}(P) \leq f_{k-1}(C_d(n)).$$

Here equality for some k with $\lfloor \frac{d}{2} \rfloor \leq k \leq d$ implies that P is neighborly.

The first fact to note is that we can restrict our attention to simplicial polytopes.

Lemma 8.24. (Klee [324] and McMullen [387]) *The vertices of a d -polytope P can be perturbed in such a way that the resulting polytope P' (with the same number of vertices) is simplicial, and*

$$f_{k-1}(P) \leq f_{k-1}(P')$$

for $0 \leq k \leq d$. Here equality for some $k > \lfloor \frac{d}{2} \rfloor$ can occur only if P is simplicial.

This is not the hard part, so we avoid the *distraction* of a proof. Thus from now on we only consider simplicial d -polytopes: this is essential, because it allows us to use the Dehn-Sommerville equations! What do they get us?

First, we note that we always have

$$f_{k-1} \leq \binom{n}{k},$$

with equality if and only if P is k -neighborly. This bound is achieved with equality for $k \leq \lfloor \frac{d}{2} \rfloor$ in the case of neighborly polytopes like the cyclic